

§8.1 - Arc Length

Problem

We want to measure the length of the curve given by

$$\{(x, f(x)) \mid x \in [a, b]\}$$

where f is a “smooth” function on $[a, b]$.

Mean Value Theorem

If f is a differentiable function on the interval $[a, b]$, then there exists a number c between a and b such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$f(b) - f(a) = f'(c)(b - a).$$

Theorem #1

Let f' be continuous on $[a, b]$.

Then the length of the curve from $(a, f(a))$ to $(b, f(b))$ is given by

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx.$$

We call L the **arc length** of f over the interval $[a, b]$.

Example #1

Find the arc length of

$$y = \frac{2}{3}x^{3/2}$$

over $[0, 3]$.

Example #2

Find the arc length of

$$y = \frac{e^x}{2} + \frac{e^{-x}}{2}$$

over $[-\ln(2), \ln(2)]$.

Definition #1

Let f' be continuous on $[a, b]$.

Then

$$s(x) = \int_a^x \sqrt{1 + (f'(t))^2} \, dt$$

for $x \in [a, b]$ is the **arc length function** of f .

Example #3

A hawk flying at 15 m/s at an altitude of 180 m accidentally drops its prey. The path of the falling prey is described by the equation

$$y = 180 - \frac{x^2}{45}$$

until it hits the ground, where y is its height above the ground and x is the horizontal distance traveled in meters.

Find the integral that measures the distance traveled by the prey from the time it is dropped until the time it hits the ground